

Web Technologies Lab – 04

<https://github.com/nzoopxw/web-tech-lab-sem5>

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Aim:

Design and implement a React.js-based menu navigation system for a university website containing menus such as About Us, Academics, Admissions, Research, Campus Life, Placements, and Contact Us. Implement dropdown menus for relevant sections and use React components, props, JSX, event handling, and React Router to provide smooth navigation between pages.

Source Code

File 1: src/components/Navbar.jsx

```
import { Link } from "react-router-dom";
import Dropdown from "../Dropdown";
import "../Navbar.css";

const menuData = [
  { label: "About Us", path: "/about" },
  {
    label: "Academics",
    items: [
      { label: "Undergraduate", path: "/academics/undergraduate" },
      { label: "Postgraduate", path: "/academics/postgraduate" },
      { label: "PhD", path: "/academics/phd" },
    ],
  },
],
```

```

},
{
  label: "Admissions",
  items: [
    { label: "Eligibility", path: "/admissions/eligibility" },
    { label: "Application Process", path: "/admissions/application" },
    { label: "Important Dates", path: "/admissions/dates" },
  ],
},
{
  label: "Research",
  items: [
    { label: "Research Areas", path: "/research/areas" },
    { label: "Publications", path: "/research/publications" },
  ],
},
{ label: "Campus Life", path: "/campus-life" },
{ label: "Placements", path: "/placements" },
{ label: "Contact Us", path: "/contact" },
];

export default function Navbar() {
  return (
    <nav className="navbar">
      <Link to="/" className="brand">University</Link>
      <ul className="menu">
        {menuData.map((menu, i) =>
          menu.items ? (
            <Dropdown key={i} label={menu.label} items={menu.items} />
          ) : (
            <li key={i} className="menu-item">
              <Link to={menu.path}>{menu.label}</Link>
            </li>
          )
        )}
      </ul>
    </nav>
  );
}

```

```

    })
  </ul>
</nav>
);
}

```

File 2: src/components/Dropdown.jsx

```

import { useState } from "react";
import { Link } from "react-router-dom";

export default function Dropdown({ label, items }) {
  const [open, setOpen] = useState(false);

  return (
    <li
      className="menu-item dropdown"
      onMouseEnter={() => setOpen(true)}
      onMouseLeave={() => setOpen(false)}
    >
      <span className="dropdown-toggle" onClick={() => setOpen((o) => !o)}>
        {label} ▼
      </span>
      {open && (
        <ul className="dropdown-menu">
          {items.map((item, i) => (
            <li key={i}>
              <Link to={item.path} onClick={() => setOpen(false)}>
                {item.label}
              </Link>
            </li>
          ))}
        </ul>
      )}
    </li>
  )}

```

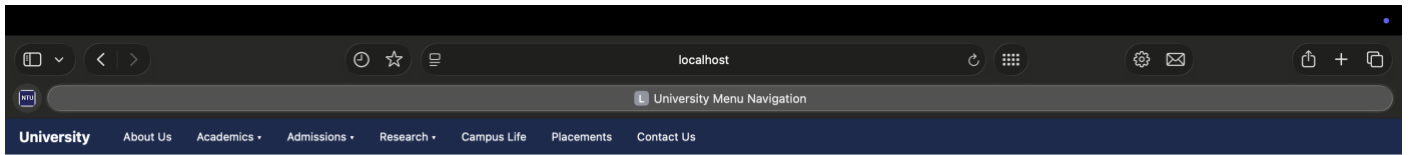
```
);
}
```

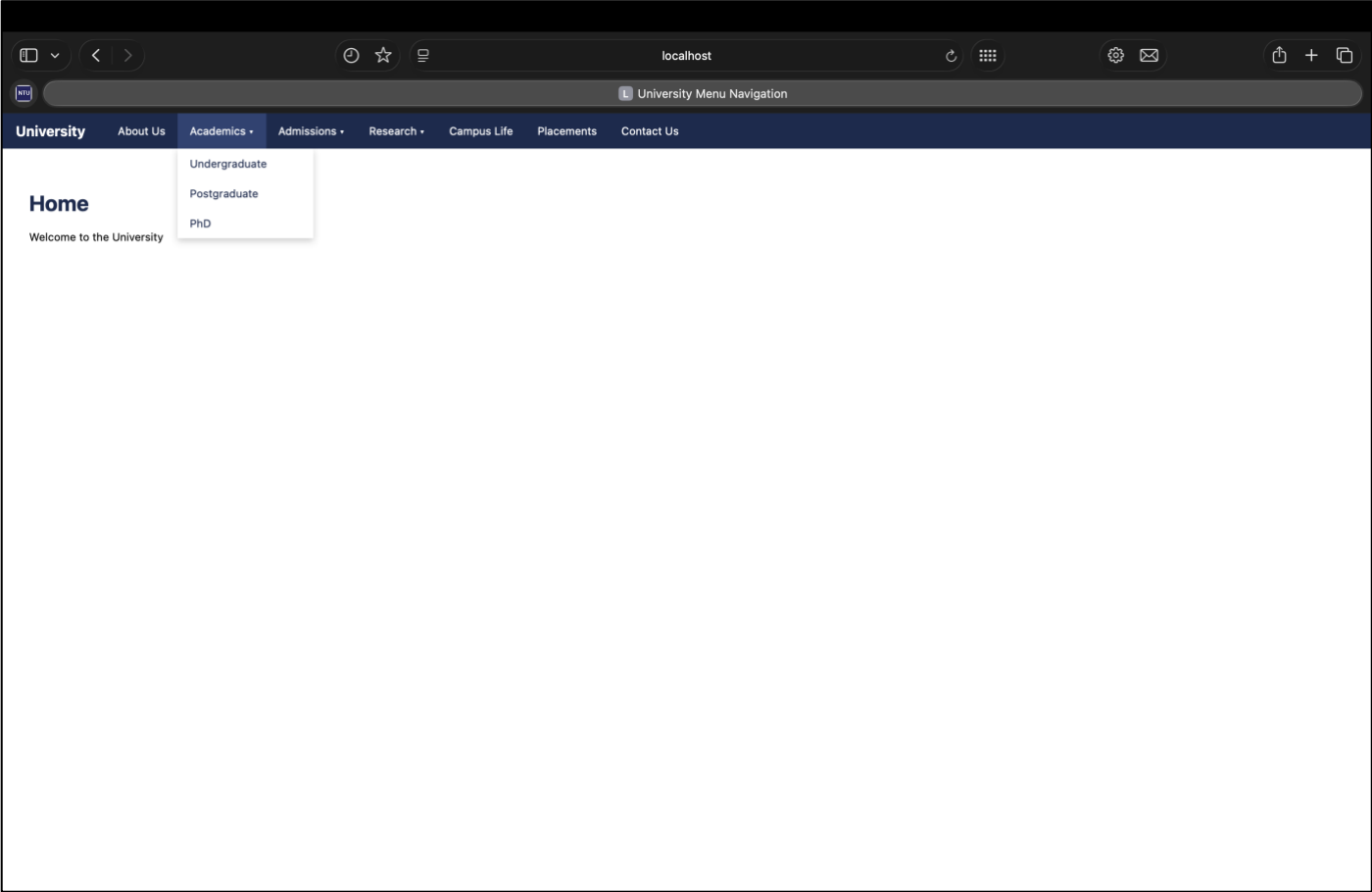
File 3: src/App.jsx

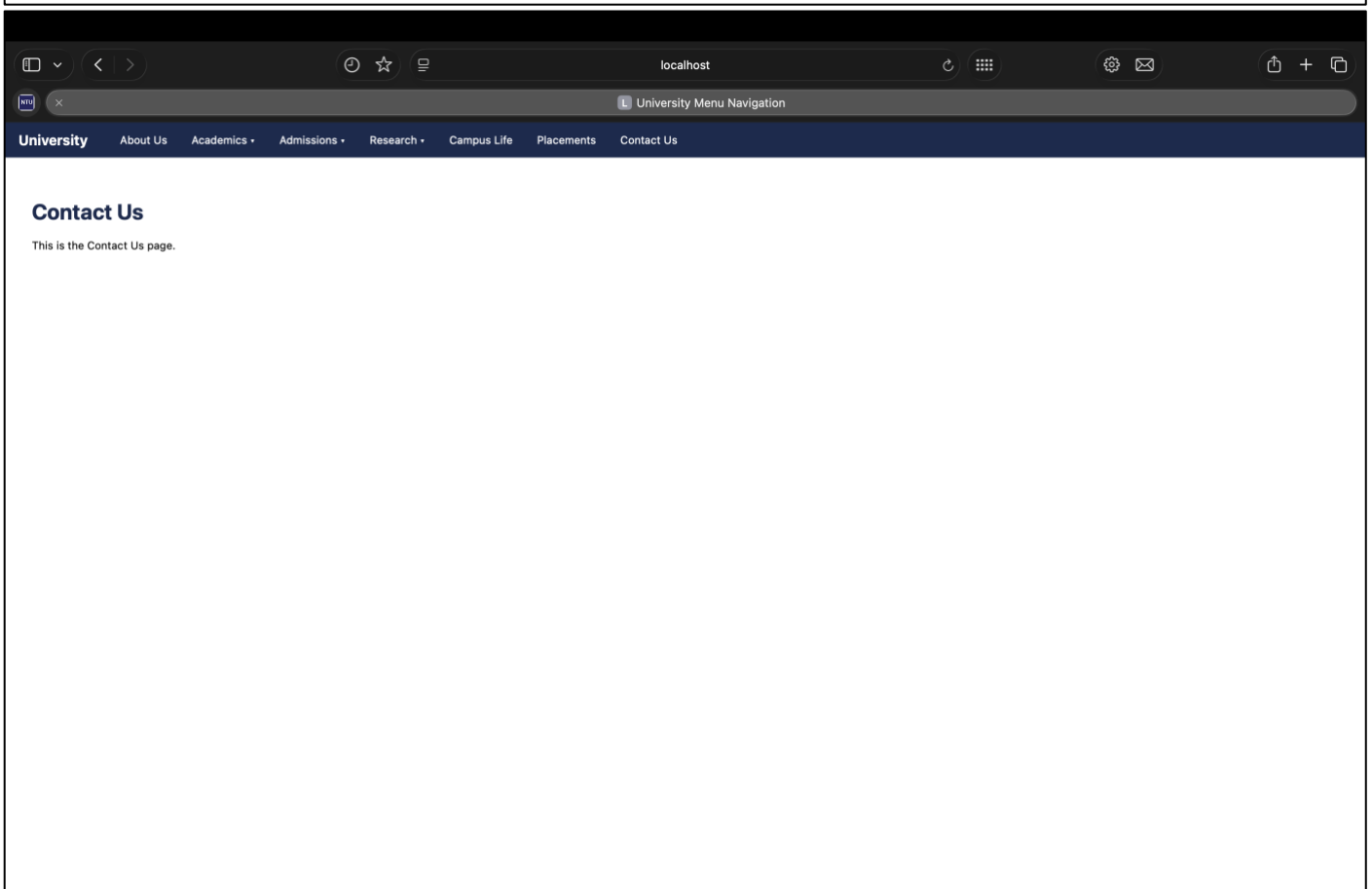
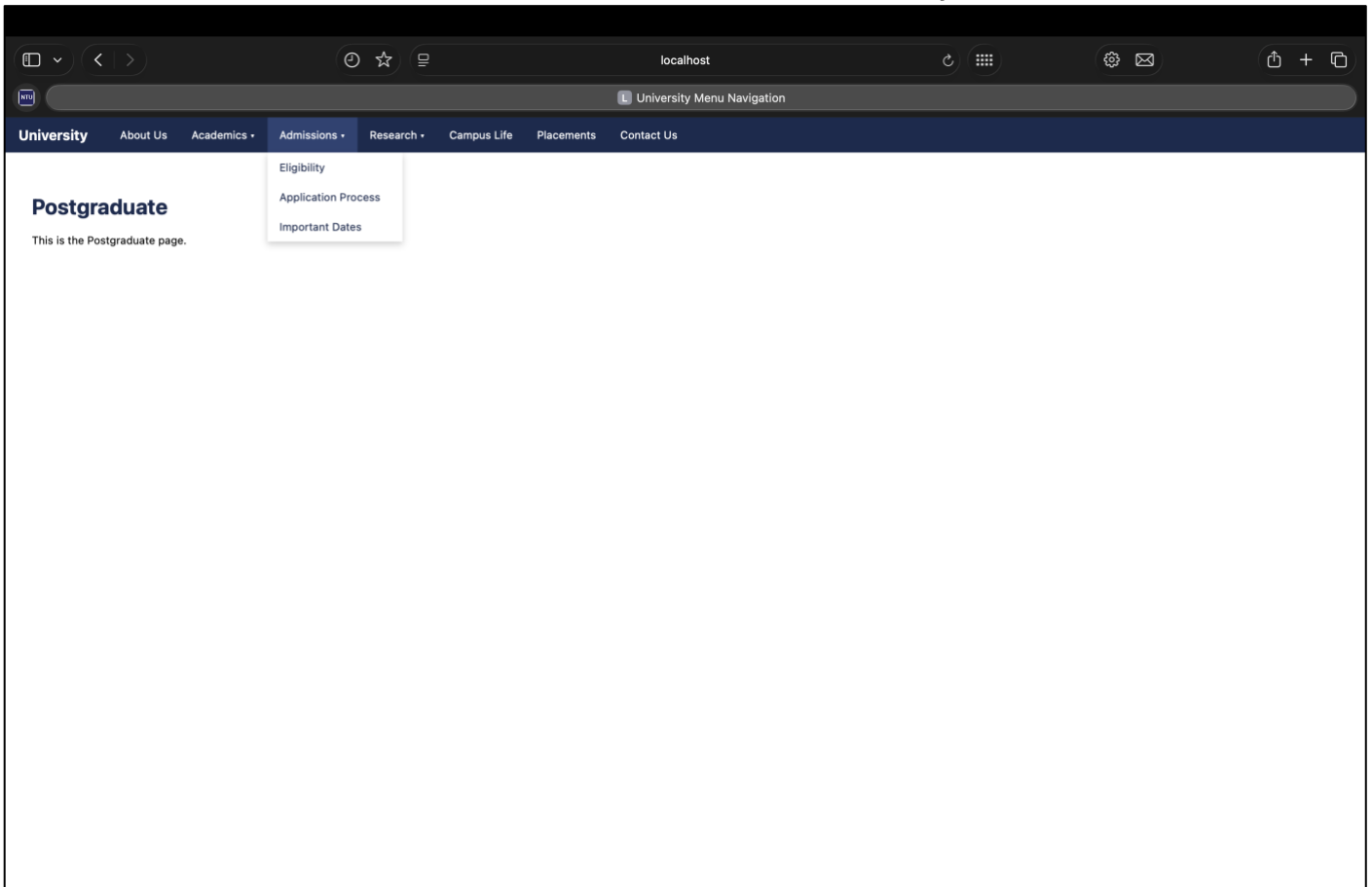
```
import { BrowserRouter as Router, Routes, Route } from "react-router-dom";
import Navbar from "../components/Navbar";
import Page from "../components/Page";
import "../App.css";

export default function App() {
  return (
    <Router>
      <Navbar />
      <div className="content">
        <Routes>
          <Route path="/" element={<Page title="Home" text="Welcome to the
University" />} />
          <Route path="/about" element={<Page title="About Us" />} />
          <Route path="/academics/undergraduate" element={<Page
title="Undergraduate" />} />
          <Route path="/academics/postgraduate" element={<Page
title="Postgraduate" />} />
          <Route path="/academics/phd" element={<Page title="PhD" />} />
          <Route path="/admissions/eligibility" element={<Page title="Eligibility" />}
/>
          <Route path="/admissions/application" element={<Page title="Application
Process" />} />
          <Route path="/admissions/dates" element={<Page title="Important Dates"
/>} />
          <Route path="/research/areas" element={<Page title="Research Areas" />}
/>
          <Route path="/research/publications" element={<Page title="Publications"
/>} />
          <Route path="/campus-life" element={<Page title="Campus Life" />} />
          <Route path="/placements" element={<Page title="Placements" />} />
          <Route path="/contact" element={<Page title="Contact Us" />} />
        </Routes>
      </div>
    </Router>
  );
}
```

Output:







Conclusion:

This assignment implements a React.js menu navigation system for a university website using reusable components, props, JSX, and event handling, with a data-driven Navbar rendering dropdown menus for sections like Academics, Admissions, and Research. React Router enables smooth client-side navigation between pages without full reloads, resulting in a clean, maintainable, and scalable navigation structure.